Arunabha Majumder

Mechanical Engineer | Mechanical Design, SolidWorks, Prototyping and Testing

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Wartsila, Gate Rudder team

Application: Mechanical Engineer, Gate Rudder (Rubbestadneset or Gdynia)

Dear Hiring Team,

I am applying for the Mechanical Engineer role in the Gate Rudder team. I am a mechanical engineer doing my PhD in Mechatronics and Robotics at Aalborg University, with my thesis expected to be finished in March. What pulls me to this role is simple: the gate rudder is about getting the same job done with less energy, and that is what I have spent the last 3 years on, just at a much smaller scale.

My main project is a hybrid shoulder exoskeleton. Instead of a big motor, I designed it so a spring carries most of the arm load and a small 6 Nm motor adds the rest. It cut energy use by more than 25% and won the Best Paper Award at IFToMM ISRM 2026. I did the design in SolidWorks from hand sketches to drawings, 3D printed and rebuilt the parts many times, and integrated the motor, encoder and force sensors. I also built custom test rigs and an Arduino-based trigger to sync a 5-bar robot with Qualisys motion capture, so I am used to proving a design with real measurements and not just on paper.

Before the PhD I worked on pneumatic artificial muscles at CSIR-CMERI, where I designed the outer PID loop of a gain-scheduled controller in MATLAB/Simulink (IEEE CONECCT 2022). In my Bachelor's I did the CAD, machining and assembly of a rehab exoskeleton myself.

I should be clear that I have no marine or ship experience yet. I would come in as a mechanical engineer who learns fast and likes to understand a product properly, from the loads to the drawings to the test results. I also have a Diploma in Fine Arts, and I often explain ideas with a quick sketch, which helps when working with different teams.

Thank you for considering my application. My portfolio is at arunabha22.github.io/arunabha-majumder.

Kind regards,
Arunabha Majumder